



Yeshwantrao Chavan College of Engineering, Nagpur

Department of Information Technology

LAB MANUAL

for

Computer Network



Nagar Yuwak Shikshan Sanstha's
YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING
(An Autonomous Institution affiliated Rashtrasanth Tukadoji Maharaj Nagpur University,

Nagpur)

HINGNA ROAD, WANADONGRI, NAGPUR

DEPARTMENT OF INFORMATION TECHNOLOGY

Academic Year: ODD 2024-2025

Course: Computer Networks Lab
Course Code: 23IT1405

Sem.:IV
Teaching Scheme: 2

Name of Faculty: Prof. Amol D Gaikwad/ Prof. W.P.Siraskar

LIST OF PRACTICALS

Exp. No	Practical List	Course Outcomes
1.	To Study Different Types of Network & Networking Commands in Linux.	CO-1
2.	To Implement Peer to Peer Connectivity by Using Crossover Cable	CO-1
3.	To Configure DNS Server using CISCO Packet Tracer	CO-1,4
4.	To Implement Client-Server Application using Java Network Programming.	CO-1
5.	Write a Program to Perform Bit Stuffing.	CO-2
6.	Write a Program to Implement CRC.	CO-2
7.	Write a Program to Implement the Hamming Code.	CO-3
8.	To Configure VPN using CISCO Packet Tracer	CO-3
9.	To Configure RIP Server using CISCO Packet Tracer.	CO-3
10.	To Configure DHCP Server using CISCO Packet Tracer	CO-4
11.	To Study Campus Networking of YCCE.	CO-1,2

Hardware and Software Requirement

Hardware Requirement

RJ-45 connector,
Crimping Tool,
Twisted pair Cable

Software Requirement

Command Prompt And Packet Tracer.

EXPERIMENT- 1

Aim: Study of basic network command and Network configuration commands.

Apparatus (Software): Command Prompt And Packet Tracer.

Procedure: To do this EXPERIMENT- follows these steps:

In this EXPERIMENT- students have to understand basic networking commands e.g ping, tracert etc.

All commands related to Network configuration which includes how to switch to privilege mode and normal mode and how to configure router interface and how to save this configuration to flash memory or permanent memory.

This commands includes

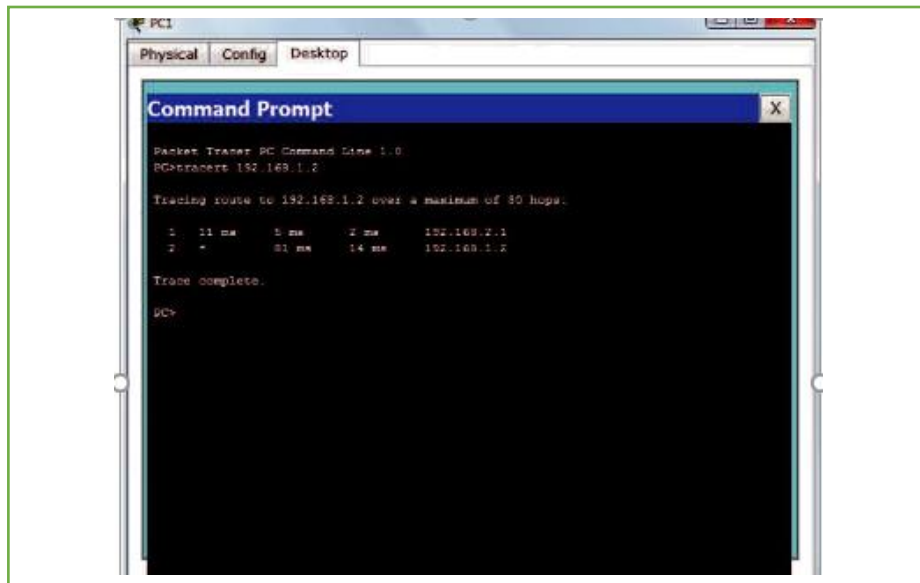
- ! Configuring the Router commands
- ! General Commands to configure network
- ! Privileged Mode commands of a router
- ! Router Processes & Statistics
- ! IP Commands
- ! Other IP Commands e.g. show ip route etc.

ping:

ping(8) sends an ICMP ECHO_REQUEST packet to the specified host. If the host responds, you get an ICMP packet back. Sound strange? Well, you can “ping” an IP address to see if a machine is alive. If there is no response, you know something is wrong.

Traceroute:

Tracert is a command which can show you the path a packet of information takes from your computer to one you specify. It will list all the routers it passes through until it reaches its destination, or fails to and is discarded. In addition to this, it will tell you how long each 'hop' from router to router takes.



nslookup:

Displays information from Domain Name System (DNS) name servers.

NOTE :If you write the command as above it shows as default your pc's server name firstly.

pathping:

A better version of traceroute that gives you statistics about packet loss and latency.

EXPERIMENT-2

Aim: Study of different types of Network cables and Practically implement the cross-wired cable and straight through cable using clamping tool.

Apparatus (Components): RJ-45 connector, Clipping Tool, Twisted pair Cable

Procedure: To do these practical following steps should be done:

1. Start by stripping off about 2 inches of the plastic jacket off the end of the cable. Be very careful at this point, as to not nick or cut into the wires, which are inside. Doing so could alter the characteristics of your cable, or even worse render it useless. Check the wires, **one more time** for nicks or cuts. If there are any, just whack the whole end off, and start over.
2. Spread the wires apart, but be sure to hold onto the base of the jacket with your other hand. You do not want the wires to become untwisted down inside the jacket. Category 5 cable must only have 1/2 of an inch of 'untwisted' wire at the end; otherwise it will be 'out of spec'. At this point, you obviously have ALOT more than 1/2 of an inch of un-twisted wire.
3. You have 2 end jacks, which must be installed on your cable. If you are using a pre-made cable, with one of the ends whacked off, you only have one end to install - the crossed over end. Below are two diagrams, which show how you need to arrange the cables for each type of cable end. Decide at this point which end you are making and examine the associated picture below.

RJ45 Pin # (END 1)	Wire Color	Diagram End #1	RJ45 Pin # (END 2)	Wire Color	Diagram End #2
1	White/Orange		1	White/Green	
2	Orange		2	Green	
3	White/Green		3	White/Orange	
4	Blue		4	White/Brown	
5	White/Blue		5	Brown	
6	Green		6	Orange	
7	White/Brown		7	Blue	
8	Brown		8	White/Blue	

Diagram shows you how to prepare Cross wired connection

EXPERIMENT-3

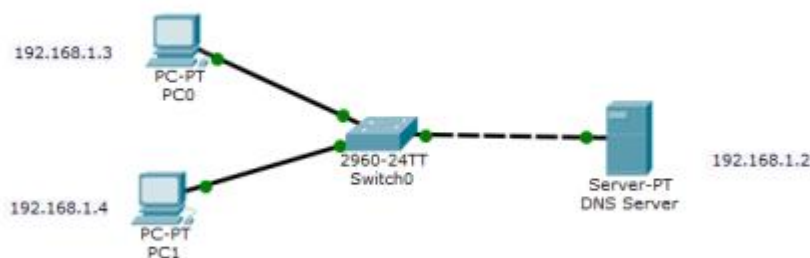
Aim: To Configure DNS Server using CISCO Packet Tracer

A **Domain Name System (DNS)** server resolves host names into IP addresses. Although we can access a network host using its IP address, DNS makes it easier by allowing us use domain names which are easier to remember. For example its much easier to access google website by typing <http://www.google.com> as compared to typing <http://208.117.229.214>. In either case, you'll access google website, but using domain name is obviously easier.

Now, before any host can use a DNS service, we must configure a DNS server first. For example, when you type the URL <http://www.google.com> in your browser, the host will query the DNS server for the IP address of <http://www.google.com>. The DNS server will resolve <http://www.google.com> into an IP address then answer back the host with the IP address.

Now let's configure a DNS server in Packet Tracer.

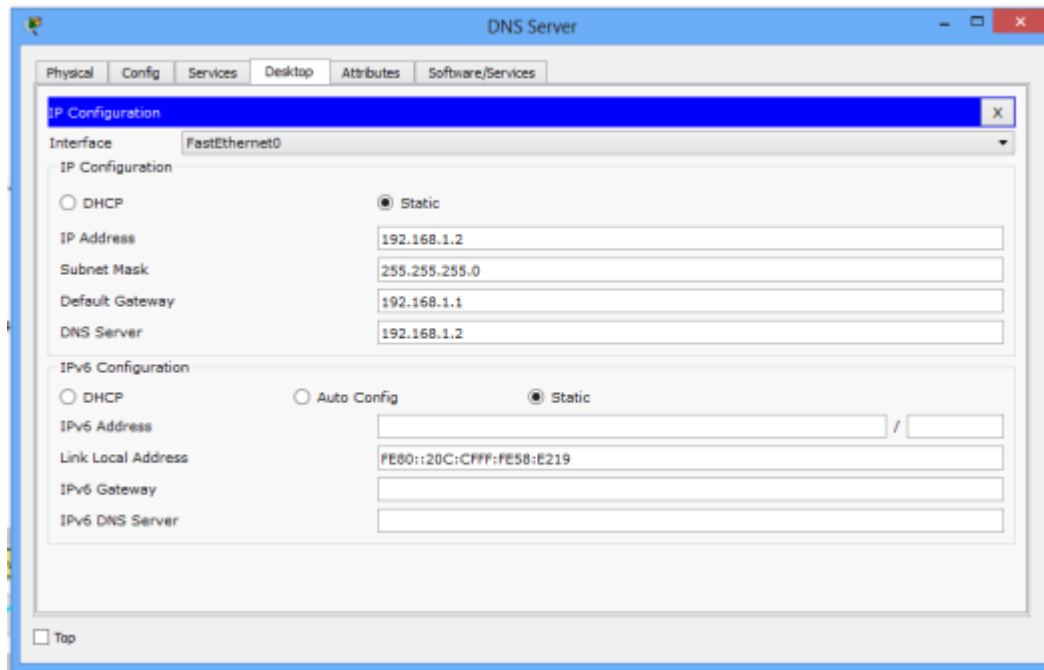
1. Build the network topology.



2. Configure static IP addresses on the PCs and the server.

Server

IP address: 192.168.1.2 **Subnet mask:** 255.255.255.0 **Default gateway:** 192.168.1.1 **DNS Server:** 192.168.1.2

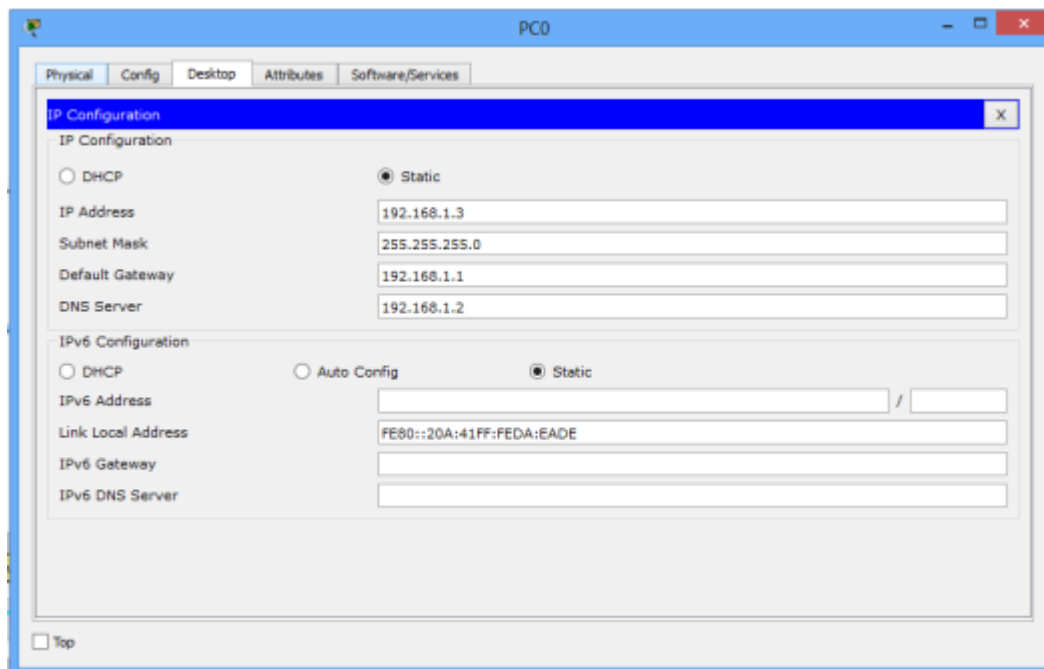


Advertisements

REPORT THIS AD

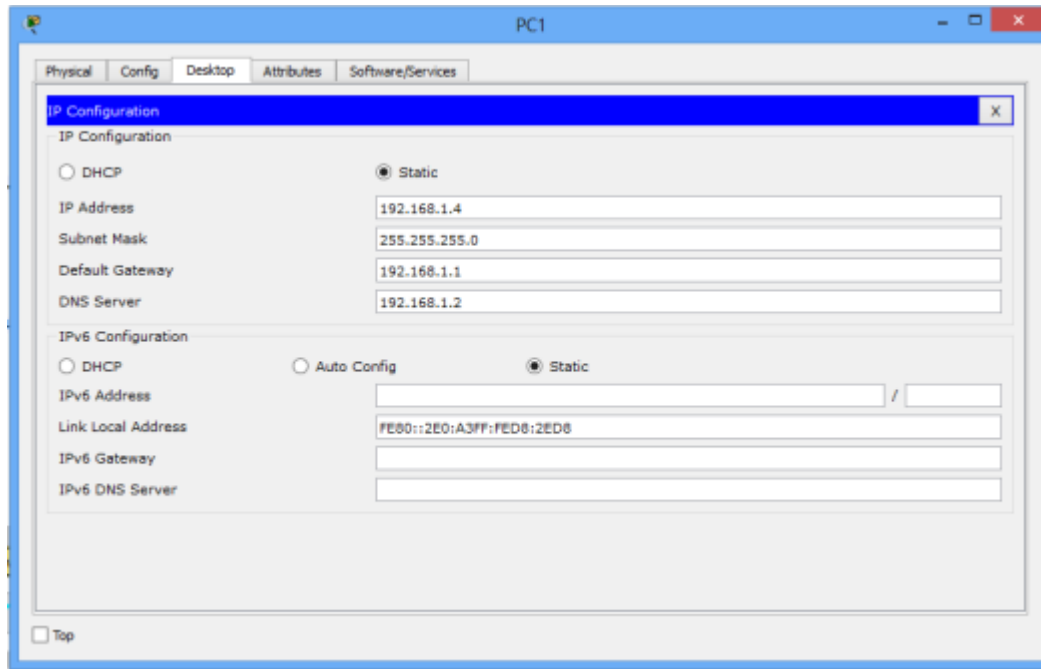
PC0

IP add: 192.168.1.3 **Subnet mask:** 255.255.255.0 **Default gateway:** 192.168.1.1 **DNS server:** 192.168.1.2



PC1

IP address: 192.168.1.4 **Subnet mask:** 255.255.255.0 **Default gateway:** 192.168.1.1 **DNS Server:** 192.168.1.2



3.Configure DNS service on the generic server.

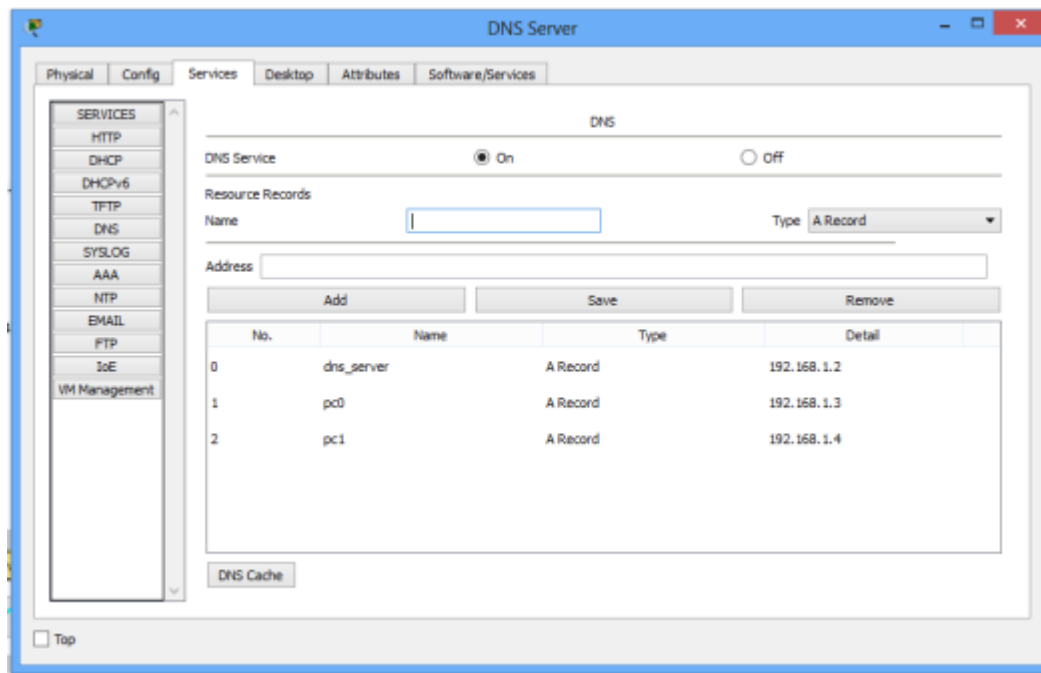
To do this, click on the server, then Click on **Services** tab. Click on **DNS server** from the menu. First turn **ON** the DNS service, then define **names** of the hosts and their corresponding **IP addresses**.

For example, to specify the DNS entry for PC0: In the **name** and **address** fields, type:

Name: PC0 **Address:** 192.168.1.3

Click on **add** then **save**. Repeat this for the PC1 and the server.

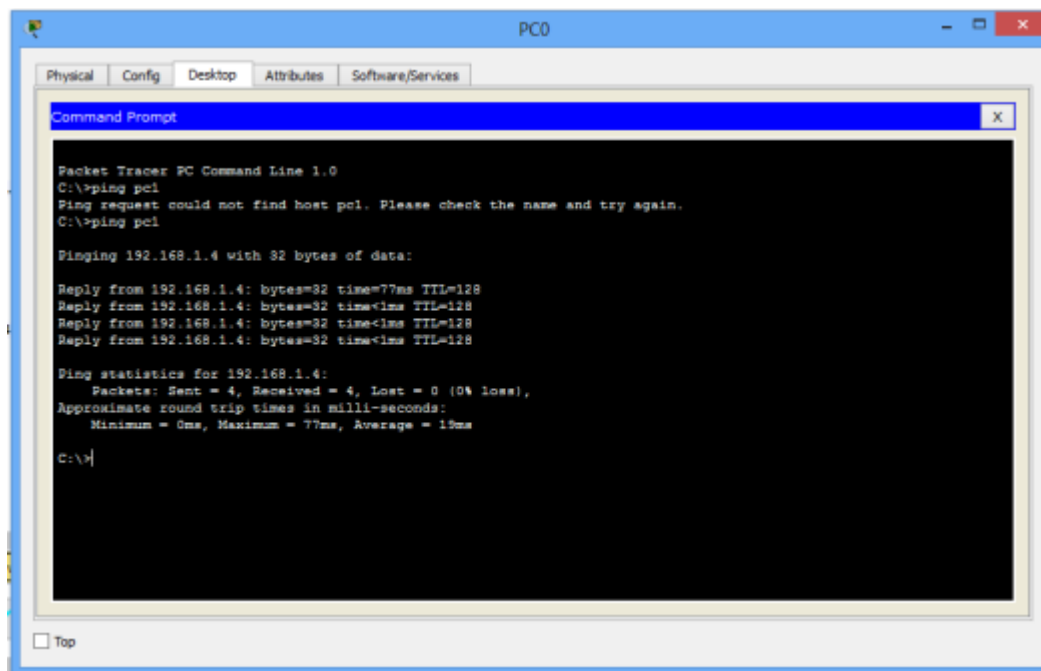
Once you're done, your DNS entries will look like this:



Finally,

4. Test **domain name – IP resolution**. Ping the hosts from one another using their names instead of their IP addresses. If the DNS service is turned on and all IP configurations are okay, then ping should work.

For example, ping PC1 from PC0. Ping should be successful.



EXPERIMENT-4

Aim: To Implement Client-Server Application using Java Network Programming.

Java Socket Programming

Creating Server:

To create the server application, we need to create the instance of `ServerSocket` class. Here, we are using 6666 port number for the communication between the client and server. You may also choose any other port number. The `accept()` method waits for the client. If clients connects with the given port number, it returns an instance of `Socket`.

1. `ServerSocket ss=new ServerSocket(6666);`
2. `Socket s=ss.accept();//establishes connection and waits for the client`

Creating Client:

To create the client application, we need to create the instance of `Socket` class. Here, we need to pass the IP address or hostname of the Server and a port number. Here, we are using "localhost" because our server is running on same system.

1. `Socket s=new Socket("localhost",6666);`

Let's see a simple of Java socket programming where client sends a text and server receives and prints it.

File: MyServer.java

1. `import java.io.*;`
2. `import java.net.*;`
3. `public class MyServer {`
4. `public static void main(String[] args){`
5. `try{`
6. `ServerSocket ss=new ServerSocket(6666);`
7. `Socket s=ss.accept();//establishes connection`
8. `DataInputStream dis=new DataInputStream(s.getInputStream());`
9. `String str=(String)dis.readUTF();`
10. `System.out.println("message= "+str);`
11. `ss.close();`
12. `}catch(Exception e){System.out.println(e);}`
13. `}`
14. `}`

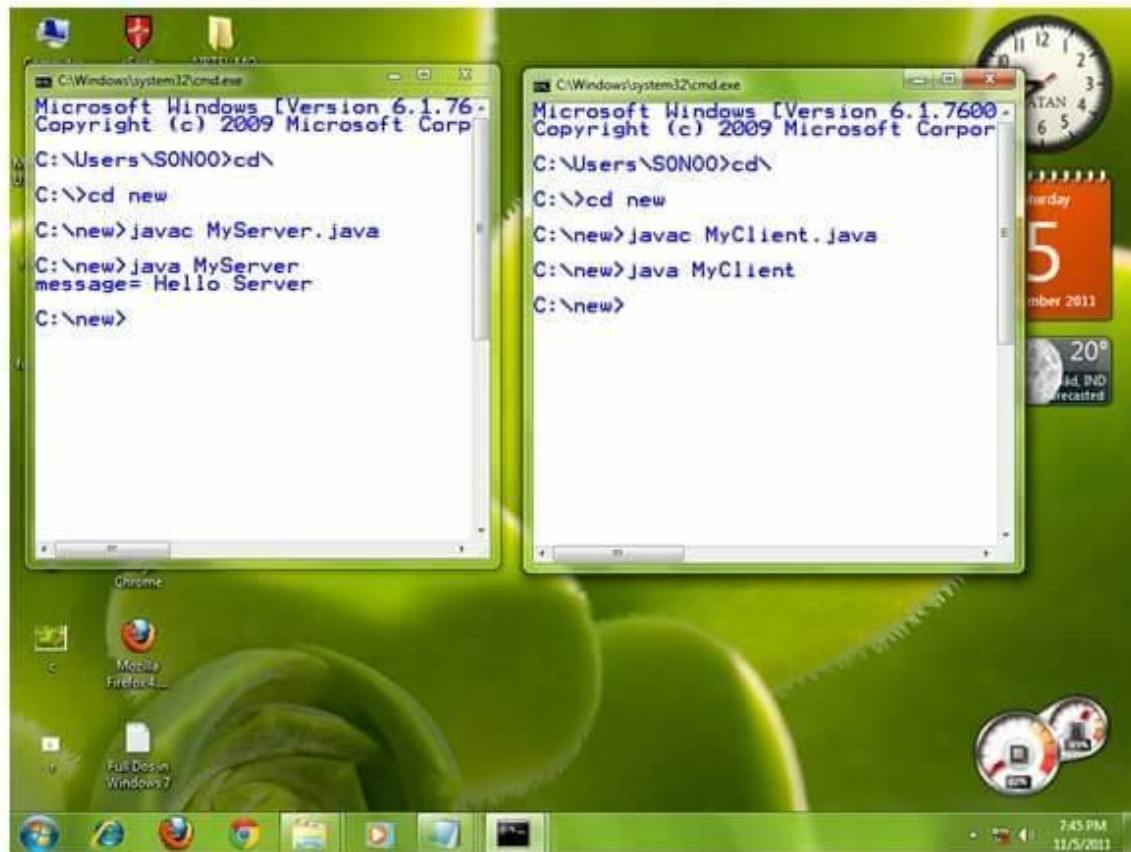
File: MyClient.java

1. **import** java.io.*;
2. **import** java.net.*;
3. **public class** MyClient {
4. **public static void** main(String[] args) {
5. **try**{
6. Socket s=**new** Socket("localhost",6666);
7. DataOutputStream dout=**new** DataOutputStream(s.getOutputStream());
8. dout.writeUTF("Hello Server");
9. dout.flush();
10. dout.close();
11. s.close();
12. }**catch**(Exception e){System.out.println(e);}
13. }
14. }

[download this example](#)

To execute this program open two command prompts and execute each program at each command prompt as displayed in the below figure.

After running the client application, a message will be displayed on the server console.



Example of Java Socket Programming (Read-Write both side)

In this example, client will write first to the server then server will receive and print the text. Then server will write to the client and client will receive and print the text. The step goes on.

File: MyServer.java

```
1. import java.net.*;
2. import java.io.*;
3. class MyServer{
4.   public static void main(String args[])throws Exception{
5.     ServerSocket ss=new ServerSocket(3333);
6.     Socket s=ss.accept();
7.     DataInputStream din=new DataInputStream(s.getInputStream());
8.     DataOutputStream dout=new DataOutputStream(s.getOutputStream());
9.     BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
10.
11.    String str="",str2="";
12.    while(!str.equals("stop")){
13.      str=din.readUTF();
14.      System.out.println("client says: "+str);
15.      str2=br.readLine();
16.      dout.writeUTF(str2);
17.      dout.flush();
18.    }
19.    din.close();
20.    s.close();
21.    ss.close();
22.  }}
```

File: MyClient.java

```
1. import java.net.*;
2. import java.io.*;
3. class MyClient{
4.   public static void main(String args[])throws Exception{
5.     Socket s=new Socket("localhost",3333);
6.     DataInputStream din=new DataInputStream(s.getInputStream());
```

```
7.  DataOutputStream dout=new DataOutputStream(s.getOutputStream());
8.  BufferedReader br=new BufferedReader(new InputStreamReader(System.in));
9.
10. String str="",str2="";
11. while(!str.equals("stop")){
12. str=br.readLine();
13. dout.writeUTF(str);
14. dout.flush();
15. str2=din.readUTF();
16. System.out.println("Server says: "+str2);
17. }
18.
19. dout.close();
20. s.close();
21. }}
```

EXPERIMENT-4

Aim: Write a Program to Perform Bit Stuffing.

Bit stuffing is a process of inserting an extra bit as 0, once the frame sequence encountered 5 consecutive 1's.

```
#include<stdio.h>
#include<string.h>
int main()
{
    int a[20],b[30],i,j,k,count,n;
    printf("Enter frame size (Example: 8):");
    scanf("%d",&n);
    printf("Enter the frame in the form of 0 and 1 :");
    for(i=0; i<n; i++)
        scanf("%d",&a[i]);
    i=0;
    count=1;
    j=0;
    while(i<n)
    {
        if(a[i]==1)
        {
            b[j]=a[i];
            for(k=i+1; a[k]==1 && k<n && count<5; k++)
            {
                j++;
                b[j]=a[k];
                count++;
                if(count==5)
                {
                    j++;
                    b[j]=0;
                }
                i=k;
            }
        }
        else
        {
            b[j]=a[i];
        }
        i++;
        j++;
    }
    printf("After Bit Stuffing :");
    for(i=0; i<j; i++)
        printf("%d",b[i]);
    return 0;
}
```

OUTPUT for BIT STUFFING:

Enter frame size (Example: 8):12

Enter the frame in the form of 0 and 1 :0 1 0 1 1 1 1 1 0 0 1

After Bit Stuffing :0101111101001

EXPERIMENT-5

Aim: Write a Program to Implement CRC.

RC or Cyclic Redundancy Check is a method of detecting accidental changes/errors in the communication channel.

CRC uses Generator Polynomial which is available on both sender and receiver side. An example generator polynomial is of the form like $x^3 + x + 1$. This generator polynomial represents key 1011. Another example is $x^2 + 1$ that represents key 101.

n : Number of bits in data to be sent
from sender side.

k : Number of bits in the key obtained
from generator polynomial.

Sender Side (Generation of Encoded Data from Data and Generator Polynomial (or Key)):

The binary data is first augmented by adding k-1 zeros in the end of the data

Use modulo-2 binary division to divide binary data by the key and store remainder of division.

Append the remainder at the end of the data to form the encoded data and send the same

Receiver Side (Check if there are errors introduced in transmission)

Perform modulo-2 division again and if the remainder is 0, then there are no errors.

In this article we will focus only on finding the remainder i.e. check word and the code word.

Modulo 2 Division:

The process of modulo-2 binary division is the same as the familiar division process we use for decimal numbers. Just that instead of subtraction, we use XOR here.

In each step, a copy of the divisor (or data) is XORed with the k bits of the dividend (or key).

The result of the XOR operation (remainder) is (n-1) bits, which is used for the next step after 1 extra bit is pulled down to make it n bits long.

When there are no bits left to pull down, we have a result. The (n-1)-bit remainder which is appended at the sender side.

CyclicRedundancyCheck.java

1. `// import required classes and packages`
2. `package javaTpoint.MicrosoftJava;`
3. `import java.util.*;`
4. `// create CRCExample class to demonstrate the working of Cyclic Redundancy Check`
5. `class CRCExample {`

```

6. // main() method start
7. public static void main(String args[]) {
8.     // create scanner class object to take input from user
9.     Scanner scan = new Scanner(System.in);
10.    // declare n for the size of the data
11.    int size;
12.    // take the size of the data from the user
13.    System.out.println("Enter the size of the data array: ");
14.    size = scan.nextInt();
15.    // declaration of the data array
16.    int data[] = new int[size];
17.    // take bits of the data from the user
18.    System.out.println("Enter data bits in the array one by one: ");
19.    for(int i = 0 ; i < size ; i++) {
20.        System.out.println("Enter bit " + (size-i) + ":");
21.        data[i] = scan.nextInt();
22.    }
23.    // take the size of the divisor from the user
24.    System.out.println("Enter the size of the divisor array:");
25.    size = scan.nextInt();
26.    // declaration of the divisor array
27.    int divisor[] = new int[size];
28.    System.out.println("Enter divisor bits in the array one by one: ");
29.    for(int i = 0 ; i < size ; i++) {
30.        System.out.println("Enter bit " + (size-i) + ":");
31.        divisor[i] = scan.nextInt();
32.    }
33.    // Divide the input data by the input divisor and store the result in the rem
    array
34.    int rem[] = divideDataWithDivisor(data, divisor);
35.    // iterate rem using for loop to print each bit
36.    for(int i = 0; i < rem.length-1; i++) {
37.        System.out.print(rem[i]);
38.    }
39.    System.out.println("\nGenerated CRC code is: ");
40.
41.    for(int i = 0; i < data.length; i++) {

```

```

42.     System.out.print(data[i]);
43. }
44. for(int i = 0; i < rem.length-1; i++) {
45.     System.out.print(rem[i]);
46. }
47. System.out.println();
48. // we create a new array that contains the original data with its CRC code

49. // the size of the sendData array with be equal to the sum of the data and
    the rem arrays length
50. int sendData[] = new int[data.length + rem.length - 1];
51. System.out.println("Enter bits in the array which you want to send: ");
52. for(int i = 0; i < sendData.length; i++) {
53.     System.out.println("Enter bit " + (sendData.length - 1) + ":");
54.     sendData[i] = scan.nextInt();
55. }
56. receiveData(sendData, divisor);
57. }
58. // create divideDataWithDivisor() method to get CRC
59. static int[] divideDataWithDivisor(int oldData[], int divisor[]) {
60.     // declare rem[] array
61.     int rem[] = new int[divisor.length];
62.     int i;
63.     int data[] = new int[oldData.length + divisor.length];
64.     // use system's arraycopy() method for copying data into rem and data arra
        ys
65.     System.arraycopy(oldData, 0, data, 0, oldData.length);
66.     System.arraycopy(data, 0, rem, 0, divisor.length);
67.     // iterate the oldData and exor the bits of the remainder and the divisor
68.     for(i = 0; i < oldData.length; i++) {
69.         System.out.println((i+1) + ". First data bit is : " + rem[0]);
70.         System.out.print("Remainder : ");
71.         if(rem[0] == 1) {
72.             // We have to exor the remainder bits with divisor bits
73.             for(int j = 1; j < divisor.length; j++) {
74.                 rem[j-1] = exorOperation(rem[j], divisor[j]);
75.                 System.out.print(rem[j-1]);

```

```

76.     }
77. }
78. else {
79.     // We have to exor the remainder bits with 0
80.     for(int j = 1; j < divisor.length; j++) {
81.         rem[j-1] = exorOperation(rem[j], 0);
82.         System.out.print(rem[j-1]);
83.     }
84. }
85. // The last bit of the remainder will be taken from the data
86. // This is the 'carry' taken from the dividend after every step
87. // of division
88. rem[divisor.length-1] = data[i+divisor.length];
89. System.out.println(rem[divisor.length-1]);
90. }
91. return rem;
92. }
93. // create exorOperation() method to perform exor data
94. static int exorOperation(int x, int y) {
95.     // This simple function returns the exor of two bits
96.     if(x == y) {
97.         return 0;
98.     }
99.     return 1;
100. }
101. // method to print received data
102. static void receiveData(int data[], int divisor[]) {
103.
104.     int rem[] = divideDataWithDivisor(data, divisor);
105.     // Division is done
106.     for(int i = 0; i < rem.length; i++) {
107.         if(rem[i] != 0) {
108.             // if the remainder is not equal to zero, data is corrupted
109.             System.out.println("Corrupted data received...");
110.             return;
111.         }
112.     }

```

```
113.         System.out.println("Data received without any error.");
114.     }
115. }
```

Output:

```
Console
<terminated> CRCEExample [Java Application] C:\Users\rastogi ji\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_64
Enter the size of the data array:
7
Enter data bits in the array one by one:
Enter bit 7:
1
Enter bit 6:
0
Enter bit 5:
0
Enter bit 4:
1
Enter bit 3:
1
Enter bit 2:
0
Enter bit 1:
1
Enter the size of the divisor array:
4
Enter divisor bits in the array one by one:
Enter bit 4:
1
Enter bit 3:
0
Enter bit 2:
1
Enter bit 1:
1
1.) First data bit is : 1
Remainder : 0101
2.) First data bit is : 0
Remainder : 1010
3.) First data bit is : 1
Remainder : 0011
4.) First data bit is : 0
Remainder : 0110
5.) First data bit is : 0
Remainder : 1100
6.) First data bit is : 1
Remainder : 1110
7.) First data bit is : 1
Remainder : 1010
101
Generated CRC code is:
1001101101
Enter bits in the array which you want to send:
Enter bit 9:
1
Enter bit 9:
0
Enter bit 9:
0
Enter bit 9:
1
Enter bit 9:
1
Enter bit 9:
0
Enter bit 9:
1
Enter bit 9:
1
```

```
Console
<terminated> CRCExample [Java Application] C:\Users\rastogi ji\p2\pool\plugins\org.eclipse.justj.openjdk.hotspot.jre.full.win32.x86_6
0
Enter bit 9:
1
Enter bit 9:
1
Enter bit 9:
0
Enter bit 9:
1
1.) First data bit is : 1
Remainder : 0101
2.) First data bit is : 0
Remainder : 1010
3.) First data bit is : 1
Remainder : 0011
4.) First data bit is : 0
Remainder : 0111
5.) First data bit is : 0
Remainder : 1110
6.) First data bit is : 1
Remainder : 1011
7.) First data bit is : 1
Remainder : 0000
8.) First data bit is : 0
Remainder : 0000
9.) First data bit is : 0
Remainder : 0000
10.) First data bit is : 0
Remainder : 0000
Data received without any error.
```

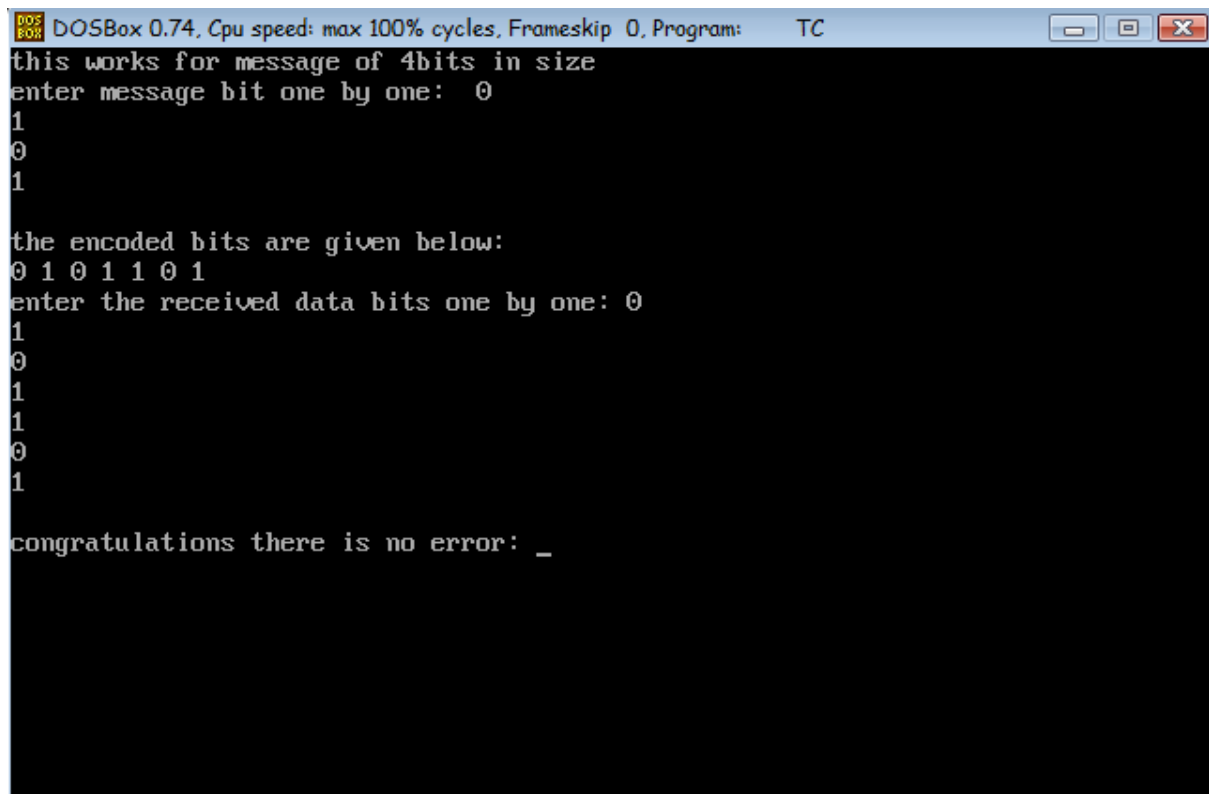
EXPERIMENT-7

Aim: Implementation of Hamming Code

C program

```
1. #include<stdio.h>
2. #include<conio.h>
3. void main() {
4.     int data[7],rec[7],i,c1,c2,c3,c;
5.     printf("this works for message of 4bits in size \nenter message bit on
   e by one: ");
6.     scanf("%d%d%d%d",&data[0],&data[1],&data[2],&data[4]);
7.     data[6]=data[0]^data[2]^data[4];
8.     data[5]=data[0]^data[1]^data[4];
9.     data[3]=data[0]^data[1]^data[2];
10.    printf("\nthe encoded bits are given below: \n");
11.    for (i=0;i<7;i++) {
12.        printf("%d ",data[i]);
13.    }
14.    printf("\nenter the received data bits one by one: ");
15.    for (i=0;i<7;i++) {
16.        scanf("%d",&rec[i]);
17.    }
18.    c1=rec[6]^rec[4]^rec[2]^rec[0];
19.    c2=rec[5]^rec[4]^rec[1]^rec[0];
20.    c3=rec[3]^rec[2]^rec[1]^rec[0];
21.    c=c3*4+c2*2+c1 ;
22.    if(c==0) {
23.        printf("\ncongratulations there is no error: ");
24.    } else {
25.        printf("\nerron on the postion: %d\nthe correct message is \n"
   ,c);
26.        if(rec[7-c]==0)
27.            rec[7-c]=1; else
28.            rec[7-c]=0;
29.        for (i=0;i<7;i++) {
30.            printf("%d ",rec[i]);
31.        }
32.    }
33.    getch();
34.}
```


Output



```
DOSBox 0.74, Cpu speed: max 100% cycles, Frameskip 0, Program: TC
this works for message of 4bits in size
enter message bit one by one: 0
1
0
1

the encoded bits are given below:
0 1 0 1 1 0 1
enter the received data bits one by one: 0
1
0
1
1
1
0
1

congratulations there is no error: _
```

reate VPN Using Cisco Packet Tracer 5.3

The following steps include project creation, configuring two routers (Router0 and Router1) to set up VPN between them, and then testing VPN connection by sending message packets from PC0 to PC1.

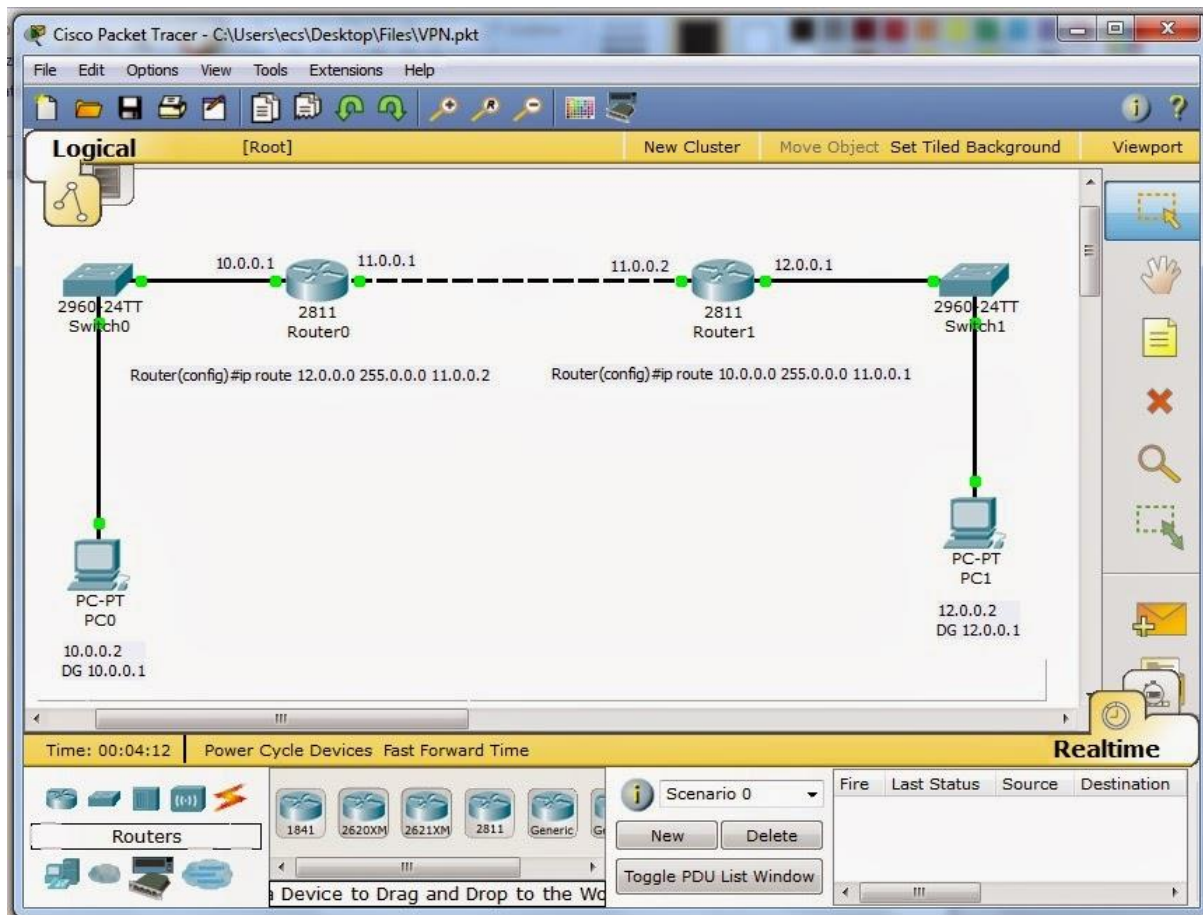
1Setup New VPN Project in CISCO Packet Tracer

At the start, you have to create a new project in packet tracer as shown in the screenshot. Attach two PC (PC0 and PC1) to switches and then connect them by placing two 2811 routers in between links.

Setup device IP address as follows

- PC0 – **10.0.0.2** attched to Router0 –**10.0.0.1** via 2960-24TT Switch0
- PC1 – **12.0.0.2** attched to Router0 –**12.0.0.1** via 2960-24TT Switch0
- Router0 – **11.0.0.1** attched to Router1 –**11.0.0.2** via Direct WAN link.

You can also download a ready-made template exactly as following [\[VPN.pkt\]](#).



New VPN Project in CISCO Packet Tracer

2Configure Router1

Configure Router1 using the following commands in the CLI environment.

Enable IPsec

```
Router(config)#crypto isakmp enable
```

Set new policy with number 1

```
Router(config)#crypto isakmp policy 1
```

Use shared key authentication method (if using certification use RSA-sig instead of pre-share)

```
Router(config-isakmp)#authentication pre-share
```

Use symmetric encryption AES

```
Router(config-isakmp)#encryption aes
```

Use hash algorithm SHA for data integrity

```
Router(config-isakmp)#hash sha
```

Use Diffie hellman group 2

```
Router(config-isakmp)#group 2
```

```
Router(config-isakmp)#exit
```

0 is the key that will be used with the next site, next site IP address 11.0.0.1, and note on packet tracer you use 0.0.0.0 instead of subnet mask

```
Router(config)#crypto isakmp key 0 address 11.0.0.1 0.0.0.0
```

Set transform-set called Yasser and esp is the protocol that will be used, you can use AH on an internal VPN

```
Router(config)#crypto ipsec transform-set yasser esp-aes esp-sha-hmac
```

Key expire after 86400 seconds

```
Router(config)#crypto ipsec security-association lifetime seconds 86400
```

ACL called Ramzy to tell which traffic will use the VPN tunnel

```
Router(config)#ip access-list extended ramzy
```

```
Router(config-ext-nacl)#permit ip 12.0.0.0 0.255.255.255 10.0.0.0 0.255.255.255
```

```
Router(config-ext-nacl)#exit
```

Create a crypto map called auda with seq number 100

```
Router(config)#crypto map auda 100 ipsec-isakmp
```

% NOTE: This new crypto map will remain disabled until a peer and a valid access list have been configured.

Link above ACL to this crypto map

```
Router(config-crypto-map)#match address ramzy
```

Link the next site IP address to this crypto map

```
Router(config-crypto-map)#set peer 11.0.0.1
```

Link DH group 2 to this crypto map

```
Router(config-crypto-map)#set pfs group2
```

Link above transform set to this crypto map

```
Router(config-crypto-map)#set transform-set yasser
```

```
Router(config-crypto-map)#ex
```

Apply crypto map auda to interface face the next site link.

```
Router(config)#int fa 0/1
```

```
Router(config-if)#crypto map auda
```

```
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

```
Router(config-if)#do wr
```

```
Building configuration...[OK]
```

```
Router(config-if)#^Z
```

```
Router#
```

3Configure Router0

Configure Router0 using the following commands in the CLI environment.

```
Router(config)#crypto isakmp enable
```

```
Router(config)#crypto isakmp policy 1
```

```
Router(config-isakmp)#authentication pre-share
```

```
Router(config-isakmp)#encryption aes
```

```
Router(config-isakmp)#group 2
```

```
Router(config-isakmp)#hash sha
```

```
Router(config-isakmp)#exit
```

```
Router(config)#crypto isakmp key 0 address 11.0.0.2 0.0.0.0
```

```
Router(config)#crypto ipsec transform-set yasser esp-aes esp-sha-hmac
```

```
Router(config)#crypto ipsec security-association lifetime seconds 86400
```

```
Router(config)#ip access-list extended ramzy
```

```
Router(config-ext-nacl)#permit ip 10.0.0.0 0.255.255.255 12.0.0.0 0.255.255.255
```

```
Router(config-ext-nacl)#exit
```

```
Router(config)#crypto map auda 100 ipsec-isakmp
```

% NOTE: This new crypto map will remain disabled until a peer

and a valid access list have been configured.

```
Router(config-crypto-map)#match address ramzy
```

```
Router(config-crypto-map)#set peer 11.0.0.2
```

```
Router(config-crypto-map)#set pfs group2
```

```
Router(config-crypto-map)#set transform-set yasser
```

```
Router(config-crypto-map)#exit
```

```
Router(config)#interface fastEthernet 0/1
```

```
Router(config-if)#crypto map auda
```

```
*Jan 3 07:16:26.785: %CRYPTO-6-ISAKMP_ON_OFF: ISAKMP is ON
```

```
Router(config-if)#exit
```

```
Router(config)#do wr
```

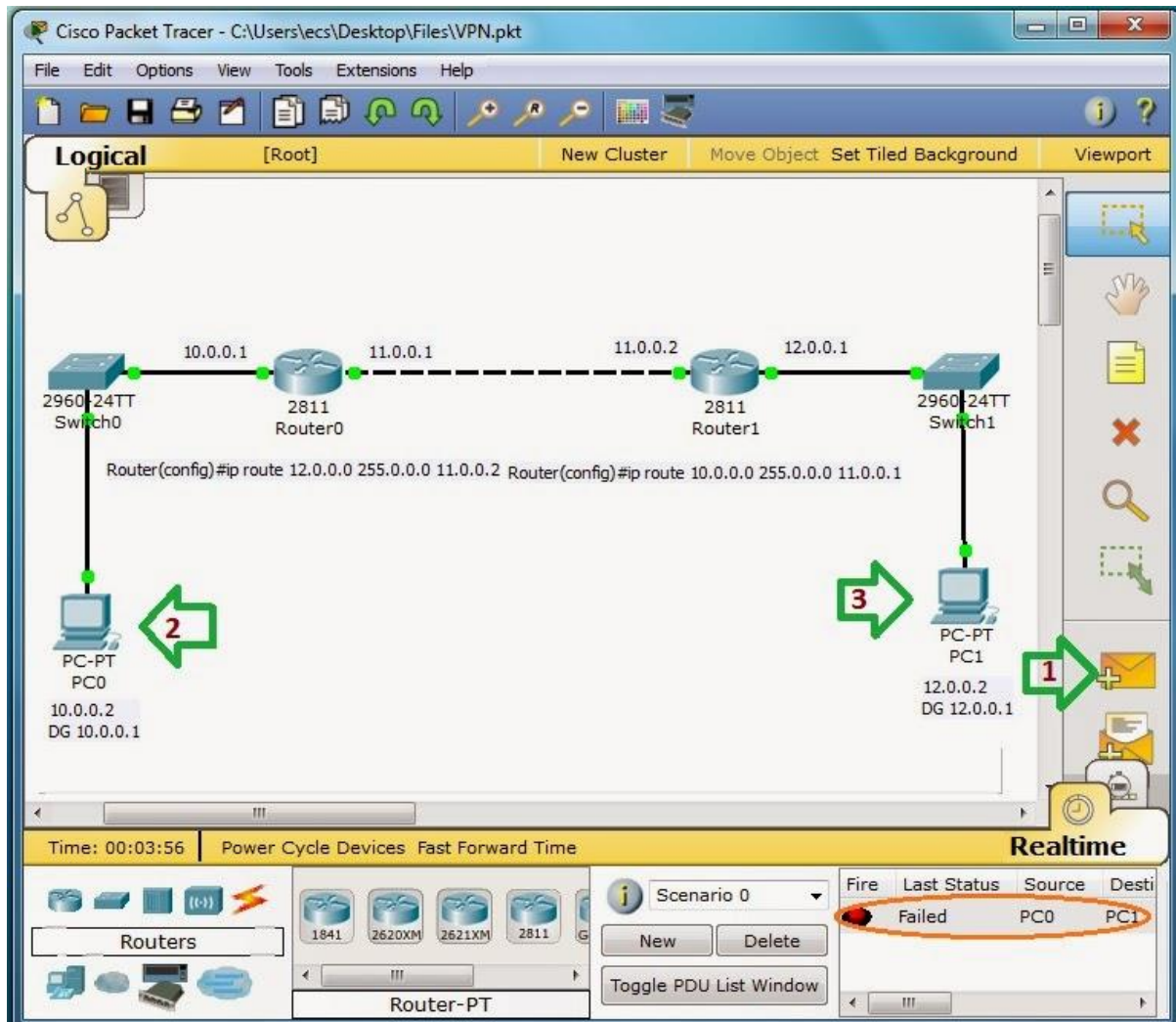
Building configuration...

[OK]

```
Router(config)#
```

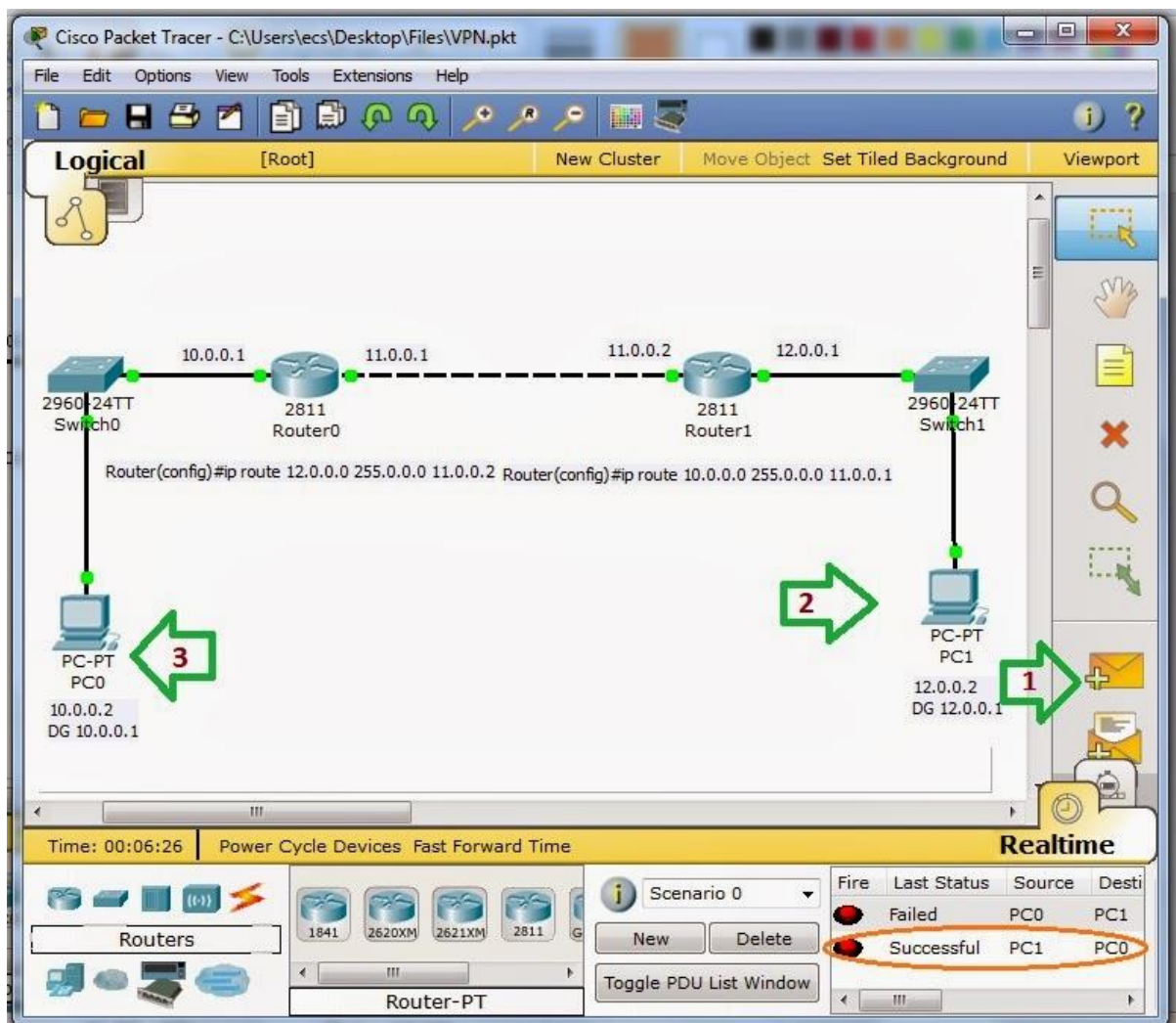
4 Test VPN Connection in CISCO Packet Tracer

To test the VPN connection send a packet from PC0 to PC1 as shown in the following figure. Don't worry message delivery will get failed because there are no IP entries on the routing table.



First-time message delivery will get failed.

Now send a packet from PC1 to PC0 & PC1 to PC0 as shown in the following figure.



This time message delivery will be successful.

This time routing table is created and it contains routing info. **So, this time message delivery will be successful.**

Fire	Last Status	Source	Destination
Failed	Failed	PC0	PC1
Successful	Successful	PC1	PC0
Successful	Successful	PC0	PC1

Final Output

To Configure RIP Server using CISCO Packet Tracer.

Routing Information Protocol (RIP) is an active routing protocol that operates hop count as a routing metric to find the most suitable route between the source and the destination network. It is a distance-vector routing protocol that has an AD value of 120 and works on the Network layer of the OSI model.

Steps to Configure and Verify Three Router Connections in Cisco Packet Tracer using RIP Routing:

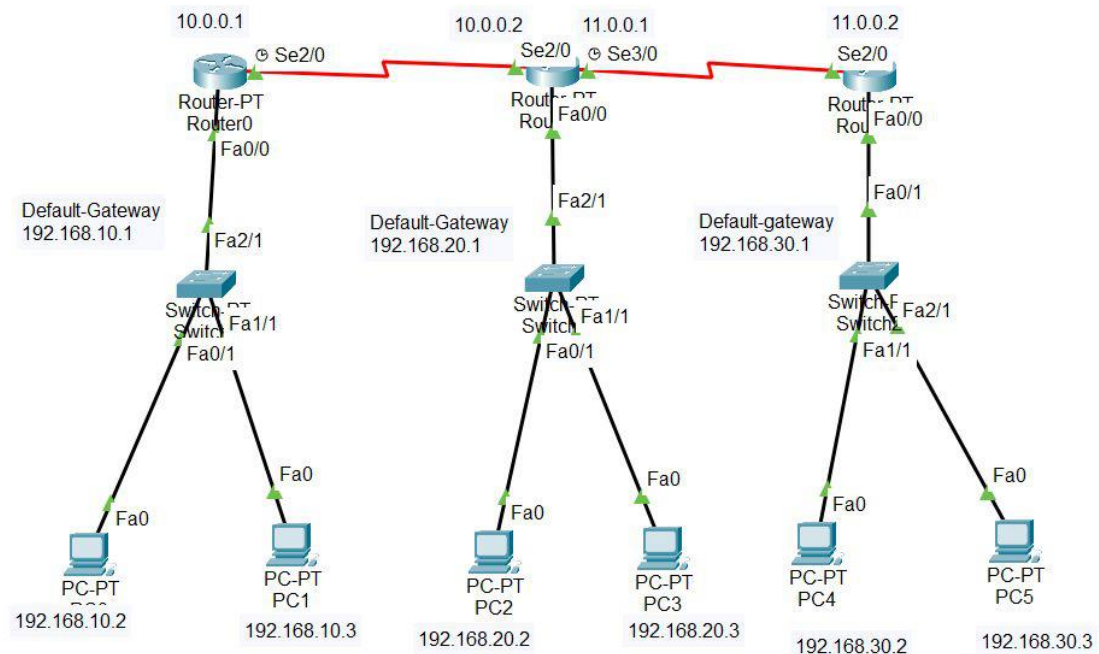
Step 1: First, open the Cisco packet tracer desktop and select the devices given below:

S.NO	Device	Model Name	Qty.
1.	PC	PC	6
2.	Switch	PT-Switch	3
3.	Router	PT-router	3

IP Addressing Table:

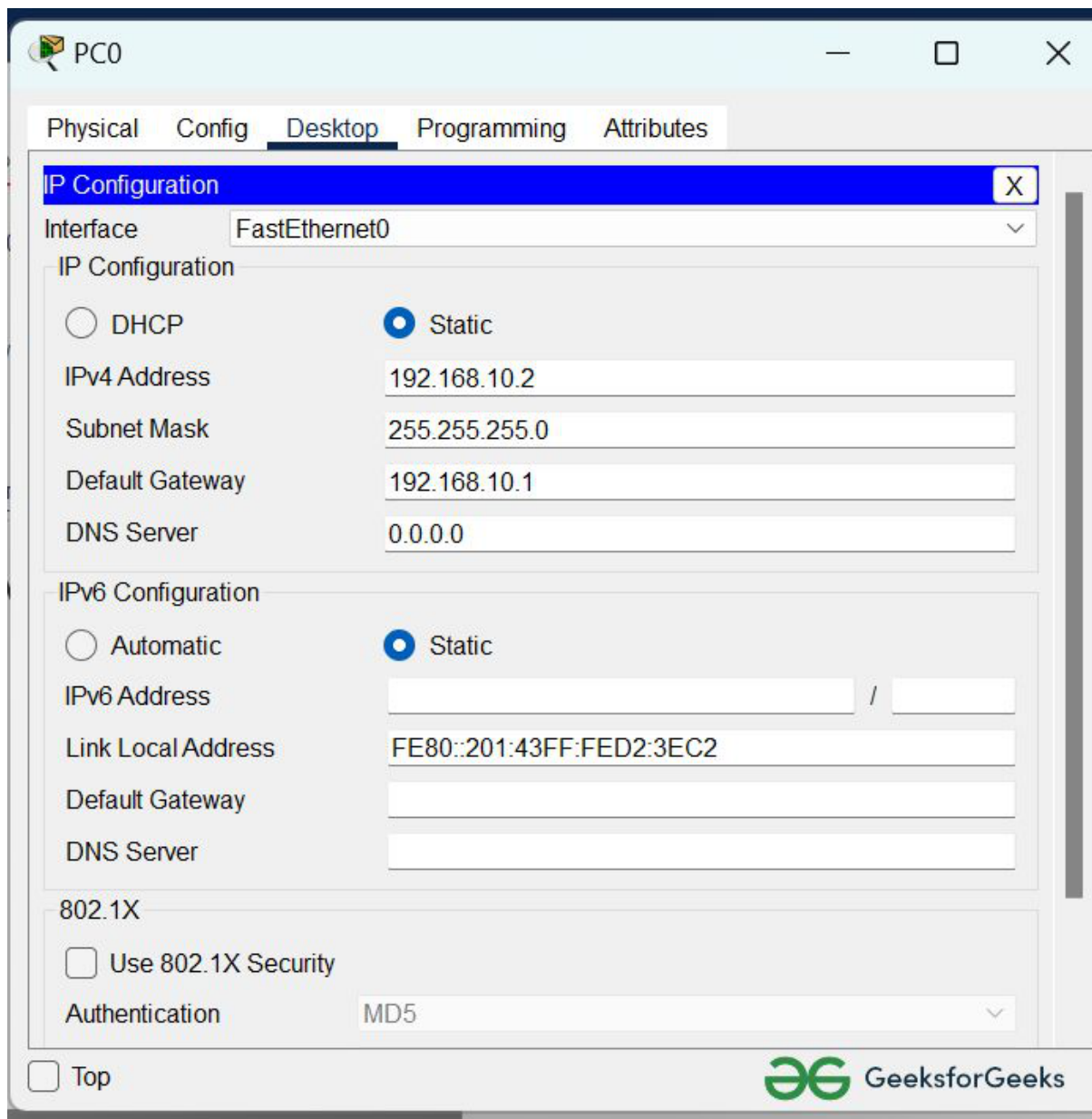
S.NO	Device	IPv4 Address	Subnet mask	Default Gateway
1.	PC0	192.168.10.2	255.255.255.0	192.168.10.1
2.	PC1	192.168.10.3	255.255.255.0	192.168.10.1
3.	PC2	192.168.20.2	255.255.255.0	192.168.20.1
4.	PC3	192.168.20.3	255.255.255.0	192.168.20.1
5.	PC4	192.168.30.2	255.255.255.0	192.168.30.1
6.	PC5	192.168.30.3	255.255.255.0	192.168.30.1

- Then, create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.



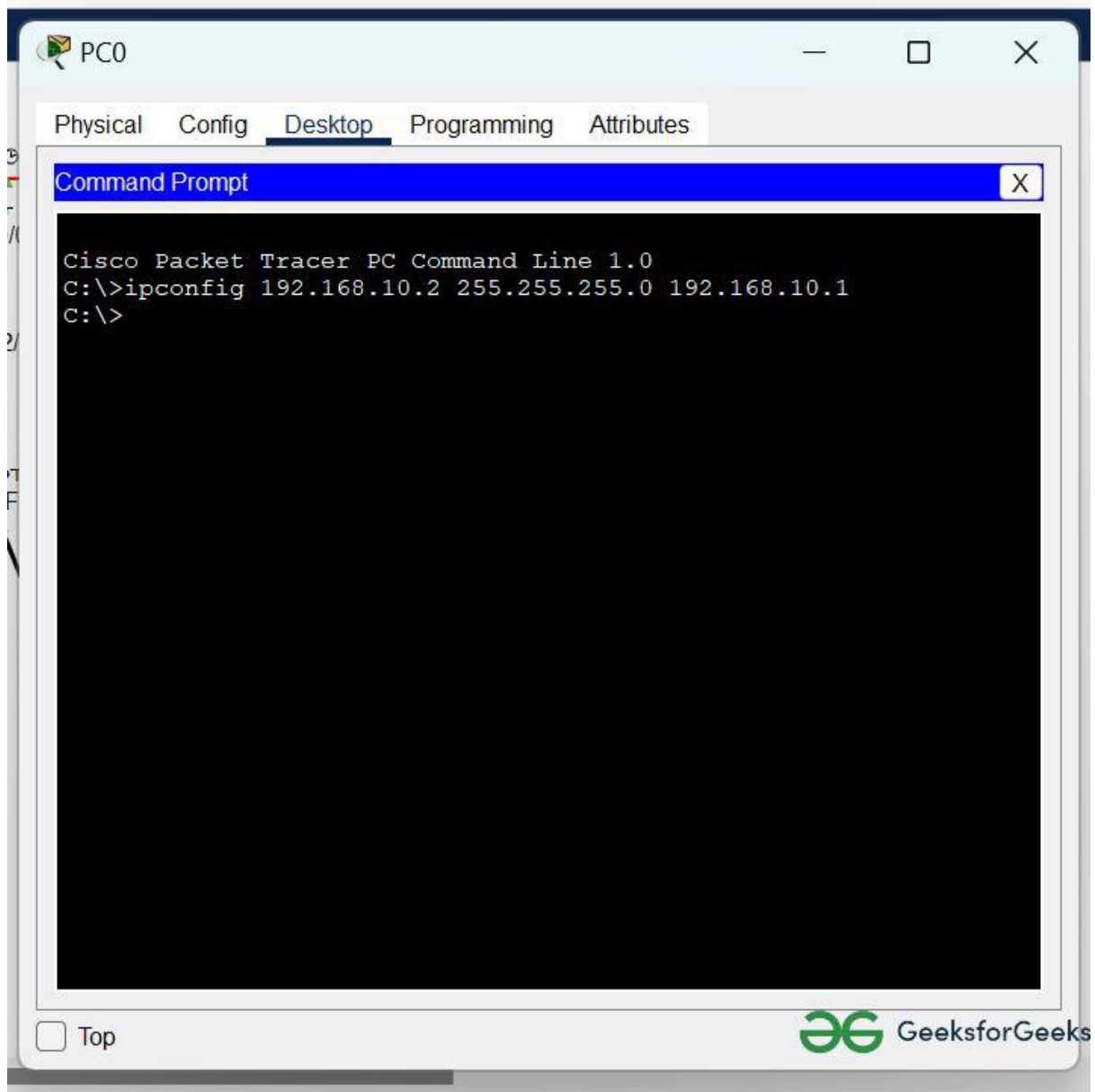
Step 2: Configure the PCs (hosts) with IPv4 address and Subnet Mask according to the IP addressing table given above.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and then IP configuration and there you will IPv4 configuration.
- Fill IPv4 address and subnet mask.



- Assigning an IP address using the ipconfig command, or we can also assign an IP address with the help of a command.
- Go to the command terminal of the PC.
- Then, type `ipConfig <IPv4 address><subnet mask><default gateway>`(if needed)

Example: `ipConfig 192.168.10.2 255.255.255.0 192.168.10.1`



- Repeat the same procedure with other PCs to configure them thoroughly.

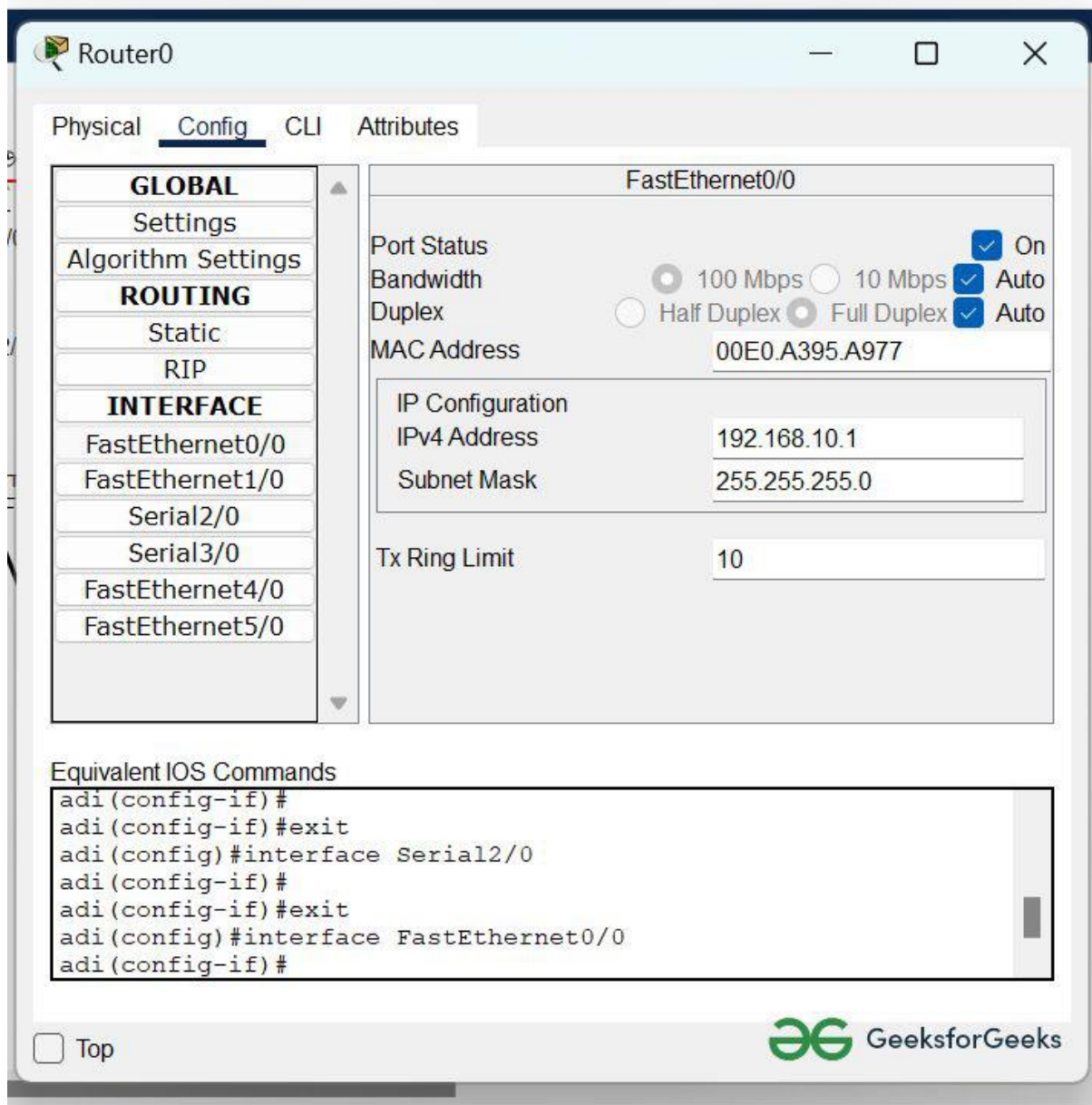
Step 3: Configure router with IP address and Subnet mask.

IP Addressing Table Router:

S.NO	Device	Interface	IPv4 Address	Subnet mask
1.	router0	FastEthernet0/0	192.168.10.1	255.255.255.0
		Serial2/0	10.0.0.1	255.0.0.0
	router1	FastEthernet0/0	192.168.20.1	255.255.255.0

S.NO	Device	Interface	IPv4 Address	Subnet mask
2.		Serial2/0	10.0.0.2	255.0.0.0
		Serial3/0	11.0.0.1	255.0.0.0
		FastEthernet0/0	192.168.30.1	255.255.255.0
3.	router2	Serial2/0	11.0.0.2	255.0.0.0

- To assign an IP address in router0, click on router0.
- Then, go to config and then Interfaces.
- Make sure to turn on the ports.
- Then, configure the IP address in FastEthernet and serial ports according to IP addressing Table.
- Fill IPv4 address and subnet mask.



- Repeat the same procedure with other routers to configure them thoroughly.

Step 4: After configuring all of the devices we need to assign the routes to the routers.

To assign RIP routes to the particular router:

- First, click on router0 then Go to CLI.
- Then type the commands and IP information given below.

CLI command : network <network id>

RIP Routes for Router0 are given below:

Router(config)#network 192.168.10.0

Router(config)#network 10.0.0.0

RIP Routes for Router1 are given below:

```
Router(config)#network 192.168.20.0
```

```
Router(config)#network 10.0.0.0
```

```
Router(config)#network 11.0.0.0
```

RIP Routes for Router2 are given below:

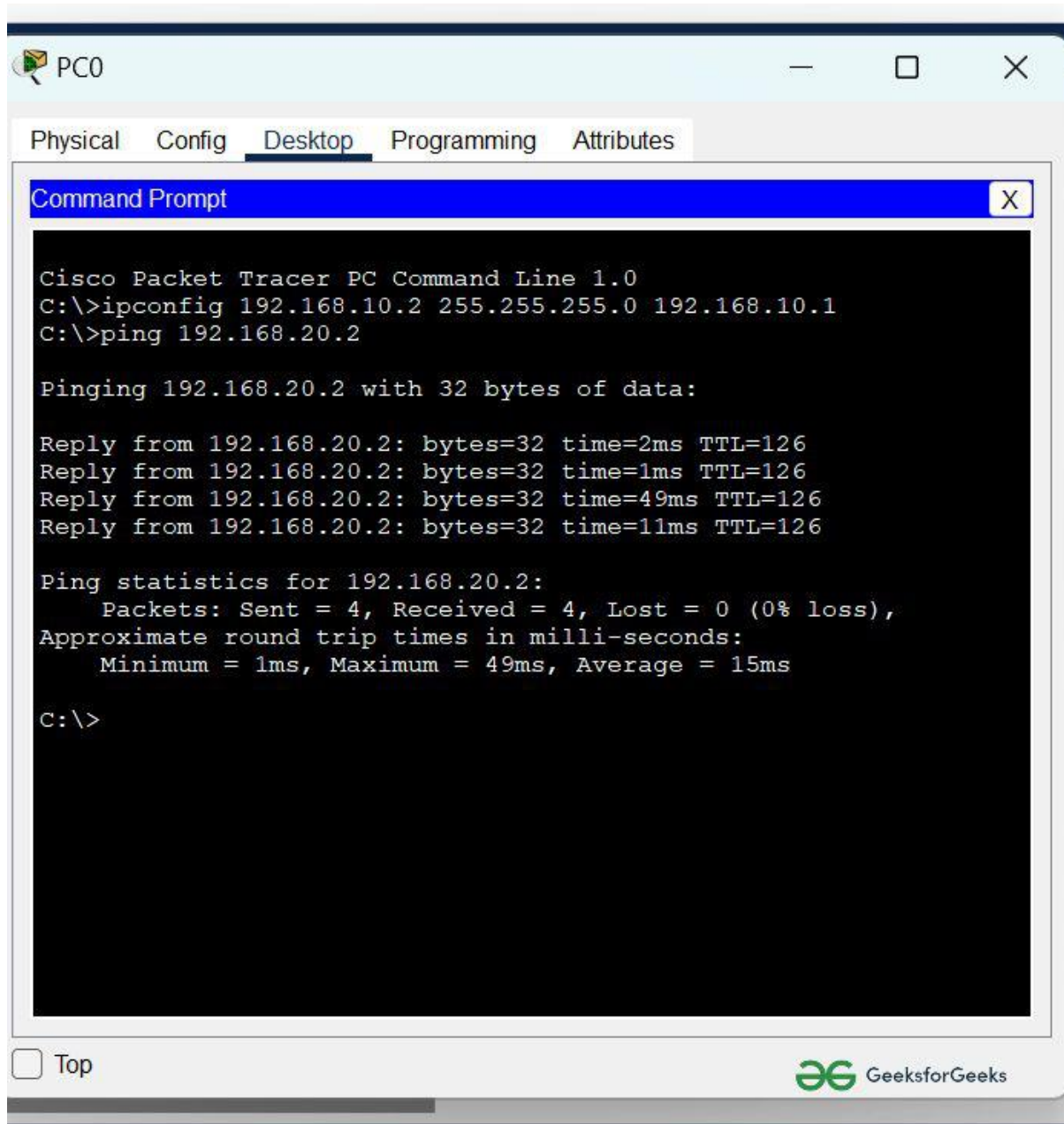
```
Router(config)#network 192.168.30.0
```

```
Router(config)#network 11.0.0.0
```

Step 5: Verifying the network by pinging the IP address of any PC.

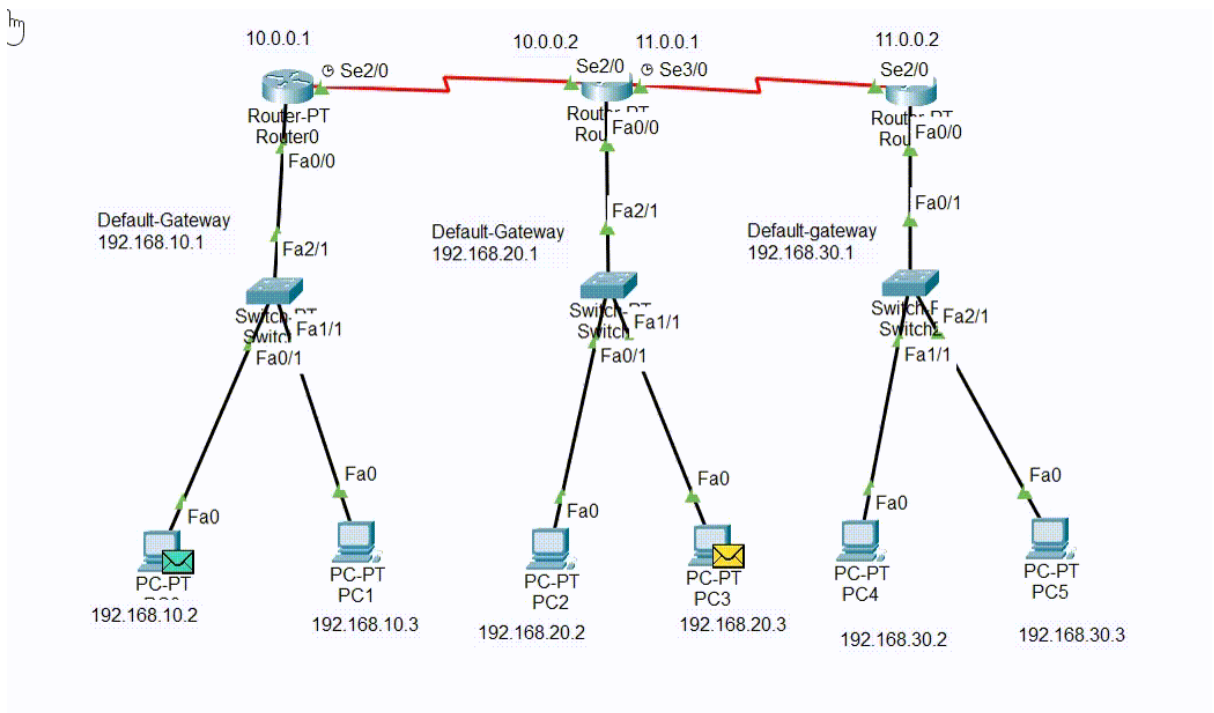
- We will use the ping command to do so.
- First, click on PC0 then Go to the command prompt.
- Then type ping <IP address of targeted node>.
- As we can see in the below image we are getting replies which means the connection is working properly.

Example : ping 192.168.20.2



- A simulation of the experiment is given below we are sending PDU from PC0 to PC2 and PC3 to PC5:

h



Experiment no 10.To Configure DHCP Server using CISCO Packet Tracer

DHCP is a network management protocol used in networks to dynamically assign IP addresses and other network configuration information like default gateway, mask, DNS server address, etc. It is an application layer protocol.

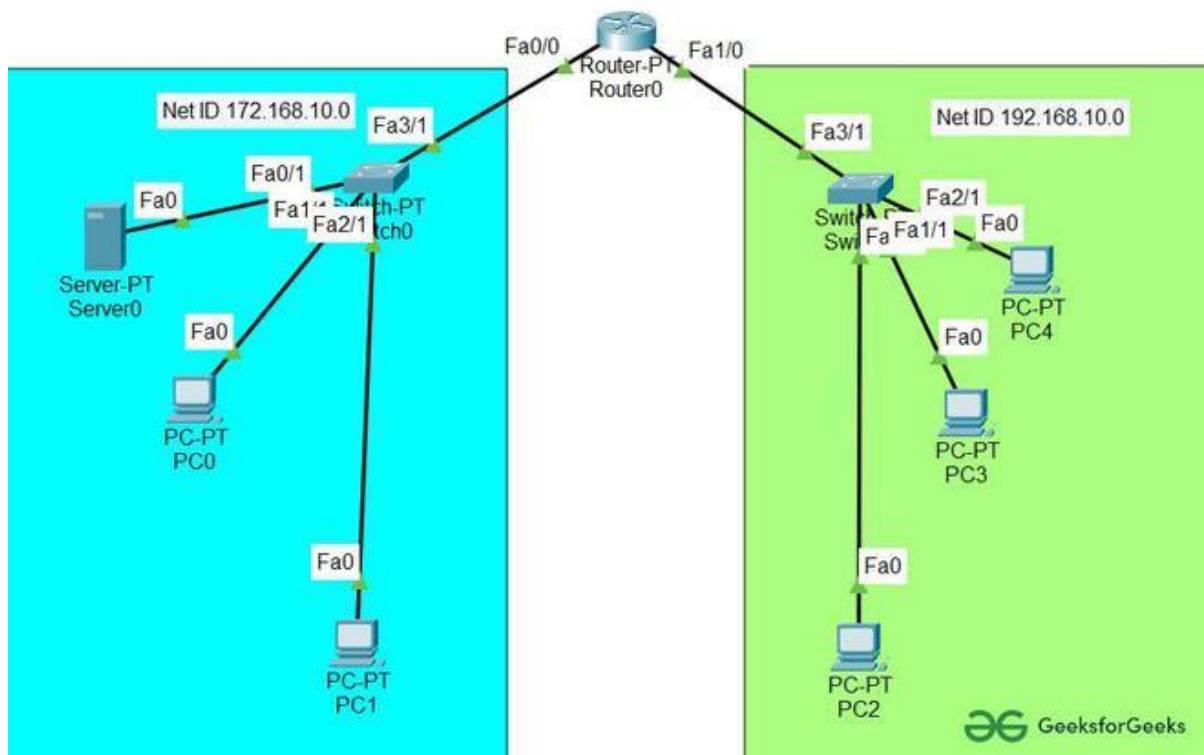
In this article, we will know about DHCP server configuration using Cisco Packet Tracer.

Steps to Configure and Verify DHCP Server in Cisco Packet Tracer:

Step 1: First, open the cisco packet tracer desktop and select the devices given below:

S.NO	Device	Model-Name	Unit
1.	PC	PC	5
2.	Switch	PT-Switch	2
3.	Router	PT-Router	1
4.	Server	Server-PT	1

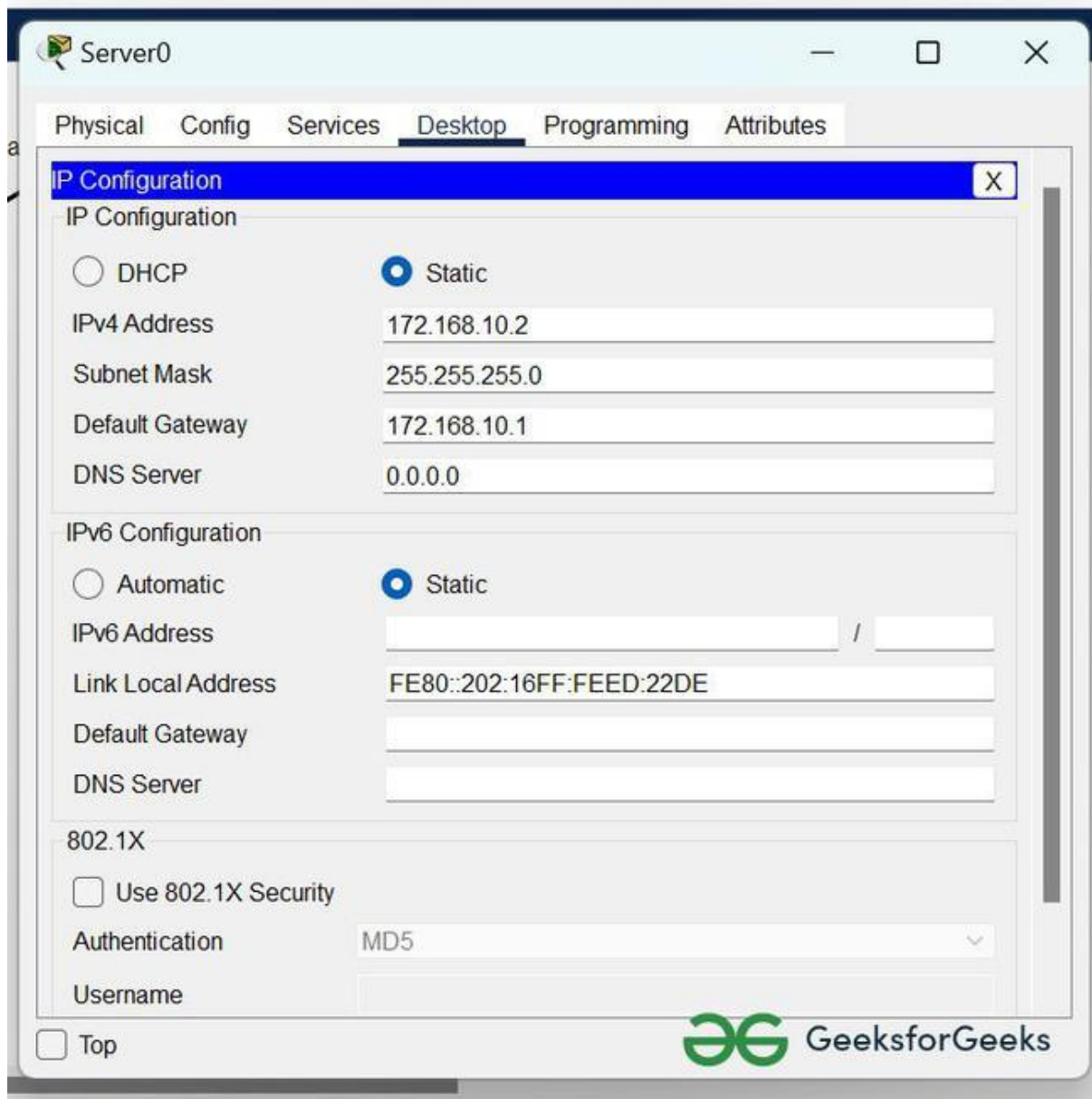
- Now create a network topology as shown below the image.
- Use an Automatic connecting cable to connect the devices with others.



Step 2: Configure the Server with IPv4 address and Subnet Mask according to the Data given above.

- To assign an IP address in Server, click on Server-PT.
- Then, go to desktop and IP configuration and there you will find IPv4 configuration.
- Add IPv4 address, subnet mask, and Default Gateway.

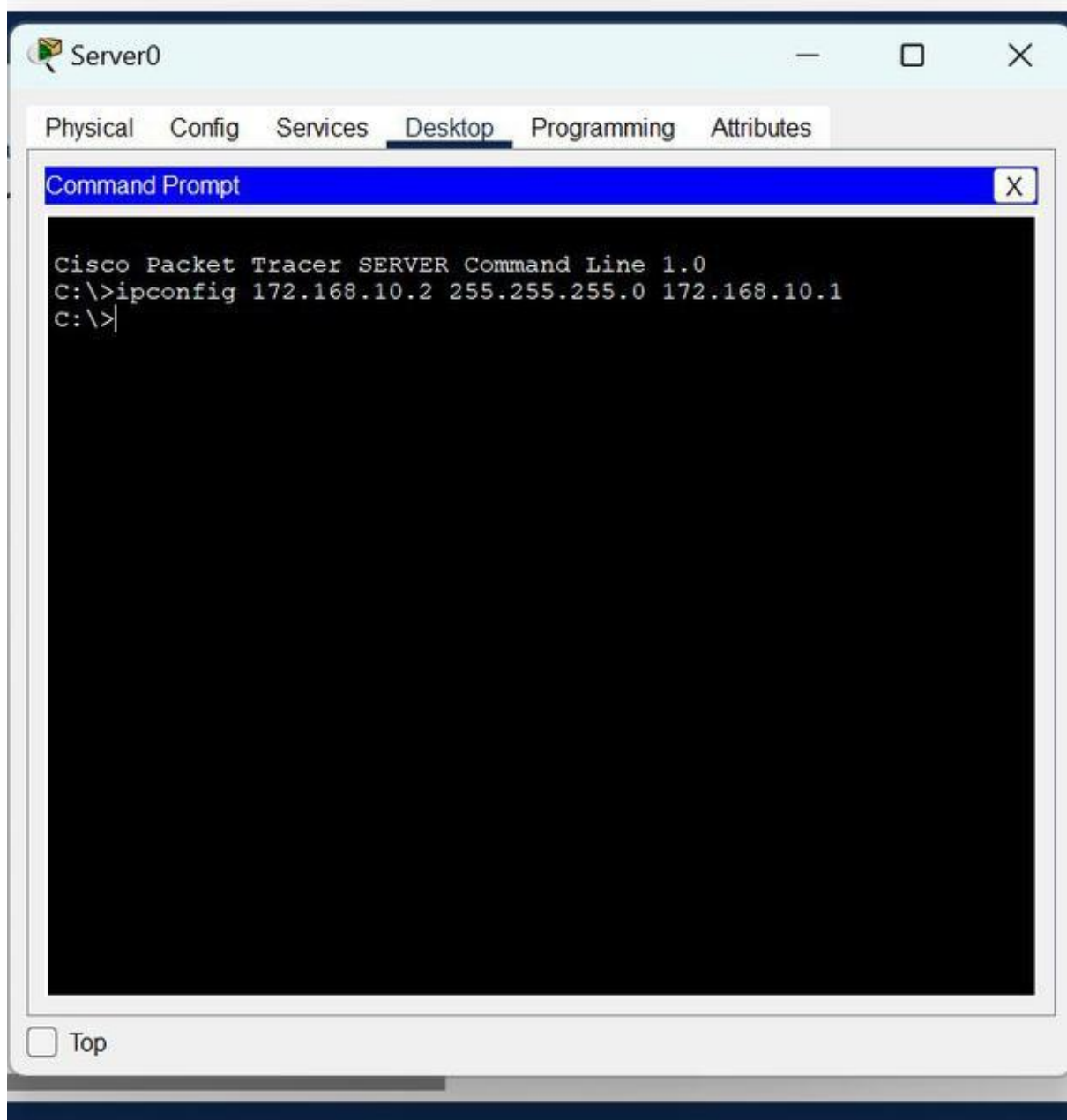
Parameters	Address value
IPv4 Address	172.168.10.2
Subnet Mask	255.255.255.0
Default-Gateway	172.168.10.1



2. Assigning IP address using the ipconfig command.

- We can also assign an IP address with the help of a command.
- Go to the command prompt of the server
- Then, type `ipconfig <IPv4 address><subnet mask><default gateway>`(if needed)

example: `ipconfig 172.168.10.2 255.255.255.0 172.168.10.1`



Step 3: Configuring the DHCP server.

To configure the DHCP server first,

- Click on Server then, Go to services.
- Click on DHCP and turn on the services and, configure the DHCP server with the help of the data given below.
 - Delete the default values of Start IP Address and subnet Mask then save the info.
 - Create two new pools.

POOL1 and POOL2 and fill the data as shown in the images below.

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES	
HTTP	
DHCP	
DHCPv6	
TFTP	
DNS	
SYSLOG	
AAA	
NTP	
EMAIL	
FTP	
IoT	
VM Management	
Radius EAP	

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Pool2

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

Start IP Addr: 192 168 10 11

Subnet Mask: 255 255 255 0

Maximum Number of Users: 255

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Server0

Physical Config **Services** Desktop Programming Attributes

SERVICES	
HTTP	
DHCP	
DHCPv6	
TFTP	
DNS	
SYSLOG	
AAA	
NTP	
EMAIL	
FTP	
IoT	
VM Management	
Radius EAP	

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: Pool2

Default Gateway: 192.168.10.1

DNS Server: 0.0.0.0

Start IP Addr: 192 168 10 11

Subnet Mask: 255 255 255 0

Maximum Number of Users: 255

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Step 4: Configuring Router with IPv4 Address and Subnet Mask.
IP Addressing Table for Router:

S.NO	Device	Interface	IPv4 Address	Subnet Mask
------	--------	-----------	--------------	-------------

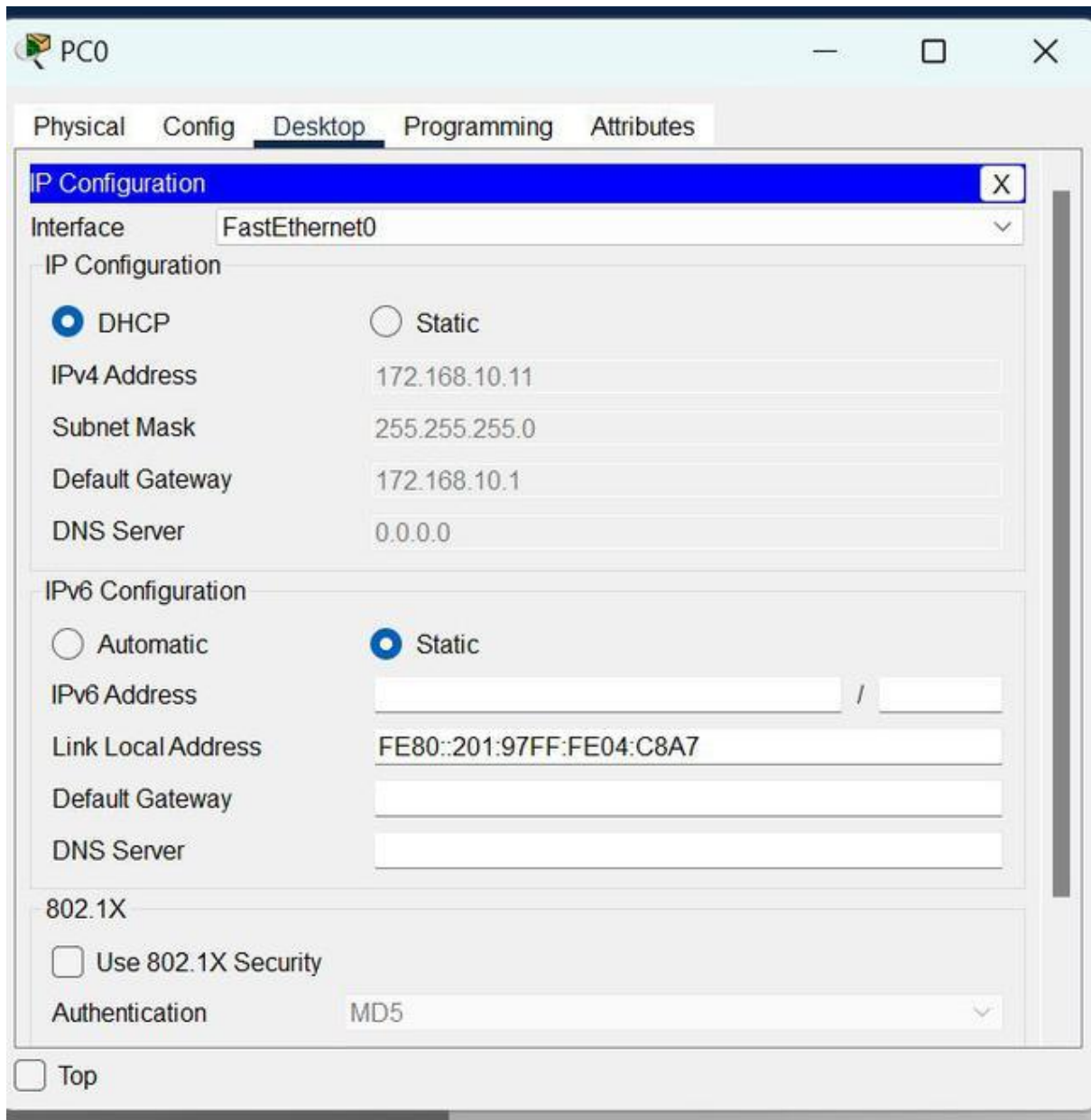
		FastEthernet0/0	172.168.10.1	255.255.255.0
--	--	-----------------	--------------	---------------

1.	router0	FastEthernet0/1	192.168.10.1	255.255.255.0
----	---------	-----------------	--------------	---------------

- To assign an IP address in router0, click on router0.
- Then, go to config and then Interfaces, and make sure to turn on the ports.
- Then, configure the IP address in FastEthernet according to IP addressing Table.
- Fill IPv4 address and subnet mask.

Step 5: Configuring the PCs and changing the IP configuration.

- To assign an IP address in PC0, click on PC0.
- Then, go to desktop and IP configuration and there you will find IPv4 configuration.
- Change its state from static to DHCP.
- It will automatically fetch the data and configure itself.



- Repeat the same procedure with other PCs to configure them thoroughly.

Output:

